

Zhangir Nurmukhambetov

Engine and Tools Programmer

+31 06 38965908 | zhangir.dev@gmail.com | jon-gear.github.io

PROFESSIONAL EXPERIENCE

Tools Developer

November 2025 – Present

SportsImproVR

Breda, Netherlands

- Built a Unity Editor one click video to FBX conversion pipeline by integrating MoveAI via a custom Python bridge, adding per clip progress and background execution, and enabling parallel conversions that cut a typical 3 video turnaround from about 15 minutes to about 5 minutes while removing manual upload, download, import steps.
- Delivered full body motion feedback on Meta Quest by bridging the native tracking gap, enabling hip and leg capture via Xsens DOT sensors. Implemented a Unity based solution that integrates the Xsens API and connects the sensors over Bluetooth on Quest, unlocking on device full body tracking where only head and hands were available.

PROJECTS

Custom Voxel-Based Engine Puzzle Platformer | C++, FMOD

May 2025 to Jun 2025

- Built a Unity style Particle System using EnTT ECS, cutting frame drops by 60%.
- Integrated 20+ FMOD sound events (e.g., footsteps, tongue, environment) for immersive feedback.
- Resolved 30+ gameplay bugs and led the Itch.io release, including trailer content and performance specs.

Custom Strategy Game Engine | C++, OpenGL, GLSL, EnTT, ImGui, Cereal

Feb 2025 to Apr 2025

- Built a real time terrain editor with brush tools and prefab support, used across 100% of game maps.
- Implemented GPU based terrain editing and tessellation rendering, reducing latency by 90%.
- Created JSON based scene serialisation with EnTT Meta, halving iteration time on level testing.

Custom Terrain Editor Tool | C++, OpenGL, GLSL, ImGui, GLM

Nov 2024 to Jan 2025

- Developed a GPU accelerated terrain sculpting tool with modular brush system and intuitive UI.
- Implemented heightmap deformation with dynamic normals and 80%+ reduction in visual latency.
- Designed foundational architecture reused in Fire Ant terrain editor, accelerating team ramp up and integration.

Custom C++ Game Engine | C++, EnTT, ImGui, Cereal, STB, tinyglTF

Sep 2024 to Nov 2024

- Built modular ECS engine with runtime editor, particle systems, scene serialisation, and GLTF support.
- Enabled real time editing, transform gizmos, and reusable asset management for future projects.
- Implemented platform abstractions for input, rendering, and file I O, supporting PC and console builds.

TECHNICAL SKILLS

Languages: C++, C#, GLSL, HLSL, Python

Frameworks & Libraries: OpenGL, EnTT (ECS), ImGui, Cereal, GLM, tinyglTF, ImGuizmo

Game Engines: Unreal Engine 5, Unity (C#), Godot

Developer Tools: Visual Studio, Git, Perforce (P4V), GitHub Actions

Languages: English (Fluent), Russian (Fluent), Arabic (Beginner)

Interests: Creative writing, Digital art and design

EDUCATION

Breda University of Applied Sciences

September 2023 to Expected Jun 2026

Bachelor of Science in Creative Media and Game Technologies

Breda, Netherlands